



**Verified Carbon
Standard**

VALIDATION REPORT

NAM HOUNG - 1 HYDRO POWER PROJECT

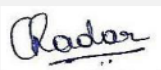


Document Prepared by:

4K Earth Science Private Limited

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Report Title	Validation report for “Nam Houg - 1 Hydro Power Project”
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Date of Issue	06-September-2024
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Summary:

A brief description of the validation and the project

Validation: 'Kosher Climate India Private Limited' has appointed '4K Earth Science Private Limited' to perform validation of the VCS project activity "Nam Houg - 1 Hydro Power Project", with regard to the relevant requirements of VCS Standard version 4.5/4/.

Project: The proposed project activity "Nam Houg - 1 Hydro Power Project" is a project which employs methodology; AMS-I.D. ver. 18.0/5/ The project activity is a hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houg - 1 Hydro Power Co., Ltd which is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of Nam Houg-1 hydro power project. Project is commissioned on 01-September-2023 the same was confirmed by verifying the commissioning certificate/17/. The project activity delivers a net annual generation of 87,280 MWh/2/.

The purpose and scope of validation

Purpose: The purpose of a validation is to have a thorough and independent assessment of the proposed project activity against the applicable VCS requirements, in particular, the project's baseline, monitoring plan and compliance with the relevant VCS requirements and host party criteria. These are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of emission reductions. 4KES's objective is to perform a thorough, independent assessment of the validation of project activity.

Scope: Validation scope is defined as an independent and objective review of the Project Description (PD). The PD is reviewed against the relevant criteria and guidance documents provided by VCS which include the following: VCS Program Guide (v4.4, dated 29/08/2023) /3/, VCS Standard (v4.5, dated 04/10/2023) /4/, Program Definitions (v4.3, dated 21/12/2022) /3/, Registration & Issuance Process (v4.3, dated 17/01/2023) /3/, VCS Validation and Verification Manual (v3.2, dated 19/10/2016) /4/ applicable at the time in order to confirm that the project meets the applicability conditions of the selected baseline and monitoring methodology AMS-I.D. ver. 18.0 /5/ also assess the claims and assumptions made in the PD without limitation on the information provided by the project participants.

The method and criteria used for validation

The validation consists of the following four phases:

- I. A desk review of the project description documents
 - A review of data and information;
 - Cross checks between information provided in PD and information from sources with all necessary means without limitations to the information provided by the project proponent;
- II. Site Visit interviews with project stakeholders
 - Interviews with relevant stakeholders in host country with personnel having knowledge with the project development via telephone, email or direct on-site visits;
 - Cross checking between information provided by interviewed personnel with all necessary means without limitations to the information provided by the project proponent;
- III. Reference to available information relating to projects or technologies similar to project under validation and review based on the approved methodology being applied for the appropriateness of formulae and accuracy of calculations.
- IV. The resolution of outstanding issues and the issuance of the final validation report and opinion.

The number of findings raised during validation

During the course of validation, a total of 05 findings were raised, which include:

02 Corrective Action Requests (CARs);

03 Clarification Requests (CLs);

00 Forward Action requests (FARs);

All the raised findings have been successfully closed by the Project Proponent.

Any uncertainties associated with the validation

There are no uncertainties associated with the validation of the project activity. The validation has been done with a reasonableness **of assumptions**.

Summary of the validation conclusion

Certification concludes the validation with a positive opinion that the VCS Project “Nam Houng - 1 Hydro Power Project” as described in the PD (version 03, dated 04/09/2024) /1/, meets all applicable VCS requirements, including those specified in the VCS Standard (v4.5) /4/, relevant methodology, tools and guidelines.

The selected baseline and monitoring methodology (AMS-I.D. ver. 18.0.)/5/ are applicable to the project and correctly applied. ‘4K Earth Science Private Limited’ Certification therefore requests the registration of the project as a VCS project.

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1 INTRODUCTION

1.1 Objective

Kosher Climate India Private Limited has commissioned '4K Earth Science Private Limited' to perform the VCS validation of the project activity: "*Nam Houng - 1 Hydro Power Project*", with regard to the relevant VCS requirements of VCS Standard version 4.5; to attain real, measurable, additional and permanent emission reductions.

The purpose of validation is to ensure a thorough, independent assessment of the project description (PD), in particular the project's description eligibility, ownership, crediting period and details regarding its compliance with the law statutes and other regulatory frameworks

- The requirements of VCS Program Guide Version 4.4 /3/
- The requirements of VCS Standard Version 4.5 /4/
- To assess the project's compliance with other relevant rules, including the project country (LAO PDR) legislation
- Other relevant rules are validated in order to confirm that the project description as documented is sound and reasonable and meet the stated requirements and identified criteria.

The validation is seen as necessary to provide assurance about the quality of the project and its intended generation of emission reductions over the project's crediting period without any double counting.

1.2 Scope and Criteria

The scope of the validation is defined as an independent and objective review of the project description document, where methodological requirements and the project's compliance in section 1 (i.e., project details) are reviewed against the requirements of VCS Standard. 4K Earth Science Private Limited has employed a risk-based approach in the validation, focusing on the identification of significant risks for project implementation and the generation of emission reductions.

The items covered in the validation are described below:

- VCS Criteria, VCS standard version 4.5 /4/
- VCS Project Description
- Background investigation and follow up interviews
- Draft validation reporting with CARs, CLs& FARs, if any
- Final validation reporting
 - The validation is based on the information made available to 4K Earth Science Private Limited and on the contract conditions.
 - The validation is not meant to provide any consulting to the project proponent. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project description. The work carried out by 4K Earth Science Private Limited is free from any conflict of interest.

1.3 Reasonableness of Assumptions

The level of assurance of the validation report falls under reasonableness of assumptions engagements as selected by the Client. The validation team verified the assumptions, limitations and methods for all the parameters and confirms that the reported ex-ante emission reductions are free from any type of material errors.

1.4 Summary Description of the Project

Nam Houng - 1 Hydro Power Project (hereafter referred to as the “the project”) is developed by Nam Houng - 1 Hydro Power Co., Ltd in Xayabouly province, Lao PDR. The primary purpose of the project is to generate electricity to the Lao Power Grid.

The project activity is a hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houng - 1 Hydro Power Co., Ltd. Project isbe commissioned on 01-September-2023 which is confirmed through verifying the commissioning certificate/17/ and the with a net annual generation of 87,280 MWh/2/.

The project is expected to constantly contribute clean energy to the Lao Power Grid. It not only conforms to the present situation, but also can adapt to future changes in the electricity market. The baseline scenario of the project is continuation of the present situation, i.e. electricity supplied from the power grid. By displacing part of the power generated by thermal power plants, the project is therefore expected to reduce CO₂ emissions by an estimation of 31,453 tCO₂e per year during the first crediting period of 7 years. The hydro project structure consists of release sluice, main powerhouse, and right-bank non overflow dam, etc. The technical specification of the equipment involved in the plant are mentioned below and the same is verified and confirmed by cross checking the equipment purchase contract/14.14/.

Turbine		
Item	Value	Unit
Number of units	3	set
Rated Capacity	5	MW
Maximum working head	22	m
Least working head	14.67	m
Rated Head	21	m
Rated speed	250	rpm
Rated flow	28.7	m ³ /s
Turbine type	Bulb	
Lifetime	30	years
Generator		
Item	Value	Unit
Number of units	3	set
Rated Capacity	5	MW
Generator Rated Power Factor	0.85	
Generator efficiency	97	%
Lifetime	30	Years

The project is not only supplying renewable electricity to grid, but also producing positive environmental and economic benefits and also, promotes the local sustainable development in the following aspects:

- During the construction period, plenty of job opportunities were provided to local residents,
- The infrastructure has been greatly improved. The enhancement of the transportation and electricity system brings substantial benefits to local villagers;
- The use of firewood will be reduced because of displacement by electricity, this reduces the damage to the local vegetation;
- Provide clean & cheap electricity in this region, promote the sustainable development in this region and slowing down the increasing trend of GHG emissions.

The Nam Houng - 1 Hydro Power development project is a part of the power development strategy by the Xayabouly Provincial Government Lao PDR to achieve the following important objectives:

- Augmenting foreign exchange earnings for Lao PDR through the sale of surplus electrical energy to consumption in the northern grid to Thailand.
- Assisting in the displacement of imported kerosene and diesel fuels by indigenous hydropower thereby reducing the outflow of foreign exchange.
- Assisting in the reduction of imported power from neighbor countries.
- Assisting in the reduction of Energy losses and stability on the EDL grid.
- Facilitating the development of agro-based and other industries.
- Improving the general standard of living of the population of Lao PDR.
- One part in Xayabouly Province has to be purchased energy by Lao Government Policy in year 2020 on 90% electrified whole country.

2 VALIDATION PROCESS

2.1 Method and Criteria

Validation was conducted using 4KES procedures in line with the requirements of VCS Standard, version 4.5 and the requirements specified in the CDM M&P, the latest version of the CDM Validation and Verification Standard version 3.0, VCS validation and verification manual version 3.2/4/ and relevant decisions of the COP/MOP and the CDM EB and applying standard auditing techniques.

The validation consisted of the following phases:

- Document review;
- On-site assessment & interviews;
- Resolution of outstanding issues
- Issuance of the validation opinion

The following sections outline each step in more detail.

2.2 Document Review

After the submission of the draft PD/1/ and supporting background documents/2/14.2/14.3/14.4/14.5/14.6/14.7/14.8/14.9/14.10/14.11/14.12/14.13/14.14/15/16/17/18 related to the project design and baseline from the client, the completeness of information made available is reviewed as per VCS standard version 4.5 requirements. Furthermore, the validation team used additional documentation like technical reports referring to the project design/1/ or to the basic conditions and technical data available on public domain/13/21/22/23/24/. The project has been listed in the VERRA (<https://registry.verra.org/app/projectDetail/VCS/4588>). A desk review is carried out to assess the project activity.

2.3 Interviews

The VVB has conducted the onsite visit for the validation of the project activity. Further the validation Team Leader along with Local Expert checked/verified on-ground information during the on-site audit. The interviews took place on 27/12/2023 and 28/12/2023. The key personnel interviewed are summarized in the table below:

S. No	Name	Role	Date	Audit Team
1	SORLAPHON	DEPUTY DIRECTOR	27/12/2023 & 28/12/2023	PURATCHIKKANAL.MA.PAA GURUBALAJI.K PHOUTKHAM SINGPHCHON (Local Expert)
2	KHAM PHOM	ADMIN MANAGER		
3	OULAVARAH SIMPMSIONS	ENVIROIMENTAL PROJECT COORDINATOR		
4	TONG KHAM	MANAGER O&M		PURATCHIKKANAL.MA.PAA GURUBALAJI.K PHOUTKHAM SINGPHCHON (Local Expert)
5	KHAM PHION	PROJECT MANAGER		
6	VIDYA KUNJAPPA	PROJECT COORDINATOR (KOSHER CLIMATE)		
7	SOUKHPHOMMA SAYADETH	LOCAL STAKEHOLDER		
8	ODATH PHOTHILATH	LOCAL STAKEHOLDER		
9	ONKEO HOMSOBATH	LOCAL STAKEHOLDER		
10	NINTASONE KHASOMPHIT	LOCAL STAKEHOLDER		

2.4 Site Visits

The VVB has conducted the site visit for the validation of the project activity. Further the validation Team Leader along with Local Expert checked/verified on-ground information during the on-site audit. The

interviews took place on 27/12/2023 and 28/12/2023. The following was carried out during the interviews:

- An assessment of the project design and technical specifications, project location, implementation status and operation of the project activity as per the VCS-PD;
- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan in the VCS-PD;
- A cross check between information provided in the VCS-PD and data from other sources such as database or similar data sources;
- A check of status and observations of monitoring practices against the requirements of the VCS-PD, the applied methodology including applicable tool(s), and, where applicable, the applied standardized baseline;
- A review of calculations and assumptions made in determining the GHG data and estimated emission reductions;
- An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

The assessment team has verified sufficient appropriate audit evidences, to reduce audit risk to an acceptably low level as requisite to achieve reasonableness of assumptions for the current validation.

2.5 Resolution of Findings

The objective of this validation is to resolve any outstanding issues which need to be clarified for 4KES's positive conclusion on the project description. To guarantee transparency a validation protocol has been customized for the project. The protocol shows in a transparent manner the requirements, means of validation and the results from verifying the identified criteria. The validation protocol consists of three tables; the different columns in these tables are described below.

A corrective action request (CAR) is raised if one of the following occurs:

- Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impair the estimate of emission reductions;
- Issues identified in a FAR during validation to be verified during verification have not been resolved by the project participants.

A clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

In summary, 03 CLs, 02 CARs, were raised during this validation which were closed successfully and details are given under Appendix 3 of this report.

2.5.1 Forward Action Requests

0 FAR is raised during the validation process.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project

The project activity is a Hydro power project with a total installed capacity of 15 MW. The project is promoted by Nam Houg - 1 Hydro Power Co., Ltd. The Project activity is a new facility (Greenfield) and the electricity generated by the project will be exported to the electricity grid. The project will therefore displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The Project Proponent plans to avail the VCS benefits for the project. In the Pre- project scenario, the entire electricity delivered to the grid by the project activity, would have otherwise been generated by the operation of grid-connected power plant. The scenario prior to the start of implementation of the project activity is generation of electricity, dominated by thermal power plants, thus leading to GHG emissions. The baseline scenario is the same as the scenario prior to the start of implementation of the project activity. The project shall result in replacing anthropogenic emissions of greenhouse gases (GHG's) estimated to be approximately 31,453 tCO₂e per year, thereon displacing 87,280 MWh/year amount of electricity from the grid over the 7 years crediting period. The hydro project structure consists of release sluice, main powerhouse, and right-bank non overflow dam, etc. The estimated amount of emission reductions per annum are 31,453 tCO₂e. The project activity falls under project category since the annual average GHG emission reductions are less than 300,000 tCO₂e.

The validation team has checked all the above-mentioned details and confirms that all the information provided is accurate.

Project proponent and other entities involved in the project

Organization name	Nam Houg - 1 Hydro Power Co., Ltd
Contact person	Phapithack Ekaphanh
Title	Director
Address	Khampheng Mueang Road (T4), Phonthan Nuea Village, Sayysathar District, Vientiane Capital

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Email	smgroad@yahoo.com info@smgroup.com.la

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna M
Title	Director
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Telephone	+91 99453 43475
Email	vamsi@kosherclimate.com narendra@kosherclimate.com

Ownership

Nam Hong - 1 Hydro Power Co., Ltd. is a SPV of Simoung Group Co., Ltd. Simoung Group Co., Ltd. is the legal owner of Nam Hong – 1 Hydro Power Project (Nam Hong - 1 Hydro Power Co., Ltd.). They have the legal right to control and operate the project activities.

The business license of the project owner are the evidences for legislative right. Besides, the project allotment license/14.13/, the equipment purchase contract/14.14/ and the power purchase agreement/14.7/ are the evidences for the ownership of the plant equipment and power generation. The same was also confirmed during the conducted interviews.

Project start date

The start date of the project activity is the commercial operation date, which is the date when the project began generating GHG emission reductions which is as per section 3.8 of the VCS standard version 4.5. Start date of project activity is 01/09/2023 which is the date of commissioning the same is confirmed by cross verifying the commissioning certificate/17/. which is considered as the start date of the project activity.

Project crediting period

Project Owner has chosen the renewable crediting period as per para 3.9.1 of the VCS standard version 4.5, with first crediting of 7 Years which starts from 01-September-2023 (Commissioning date) to 31-August-2030 with an option of two times of renewable up to 21 years. Hence the proposed crediting period is from 01-September-2023 and ends on 31-August-2030. The validation team confirms that the project crediting period is reasonably determined.

Project scale and estimated GHG emission reductions or removals

The estimated volume of emission reduction from the project activity is less than 300,000 tons of CO₂e per year. The validation team confirms that the project activity falls under the category 1 'Projects' as per para 3.10.1 of the VCS standard version 4.5, as it has the potential to reduce GHG emission less than 300,000 tCO₂e/year.

The estimated annual average and the total CO₂e emission reduction by the project activity over the renewable crediting period of 7 years are expected to be 31,453 tCO₂e and 220,174 tCO₂e respectively. The validation team confirmed that emission reduction forecast is reasonable if the underlying assumptions do not change.

Project location

The project is located in the Xayabouly Province, Lao PDR. The project boundary is defined by the following GPS coordinates and is found to be consistent. Further, project GPS coordinates as provided in the VCS PD and which are presented below, have been verified through the means of google earth software.

Geo-coordinates for project location:

Project Location	Latitude	Longitude
Xayabouly Province, Lao PDR	19° 11'19.7"N	101° 48'17.8"E

Conditions prior to project initiation

The project is a Greenfield hydro power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. Thus prior to project initiation, there was nothing at site.

In the absence of project activity, the continuation of current practice i.e., generation of equivalent amount of electricity would have been generated from grid connected fossil fuel dominated power plants. Thus, for project activity baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

Electricity Generation:

In the absence of the project activity, the electricity supply were generated mainly from grid connected power plants, where the use of fossil fuel would also lead to GHG emissions.

Project compliance with applicable laws, statutes and other regulatory frameworks

Overall, the project clearly adhere to the pertinent laws and regulations governing the development and commissioning of a hydropower project in Laos. This compliance has been cross-verified with the relevant laws, statutes, and regulatory frameworks/13/22/23/24/ and confirmed by the validation team. Below is the validation team's assessment of the project's compliance based on the provided information: Law on Electricity (amended version) No. 03/NA, dated 20/12/2011/22/:

Compliance: The project has signed a Power Purchase Agreement (PPA)/14.7/ with Electricite du Laos, a state-owned enterprise, which serves as an approval for electricity connection with the grid. This indicates adherence to the requirements outlined in the law regarding electricity generation and

distribution. The electricity generation and distribution from the project activity is conformed through the electricity license/14.6/ provided by

Environmental Protection Law (2013 Ed.) (Law No. 29/NA of 2012)/23/:

Compliance: The project has obtained an Environmental clearance certificate from the Department of Natural Resources and Environment. This clearance is based on the submission of an Initial Environmental Examination (IEE) report/14.5/, confirming the project's compliance with environmental protection laws and regulations. This demonstrates the project's commitment to environmental sustainability and adherence to legal requirements.

Decree on Environmental Impact Assessment dated January 31, 2019/24/:

Assessment: The project activity has undergone an environmental impact assessment, as evidenced by the IEE report/14.5/. The validation team cross-verified the requirements of the Decree on Environmental Impact Assessment dated January 31, 2019, with the EIA report and confirmed that the EIA report meets the requirements specified in the decree.

National Policy on Environmental and Social Sustainability of the Hydropower Sector in Lao PDR/13/:

Assessment: The project activity has undergone an environmental impact assessment, as evidenced by the IEE certificate/14.5/. The validation team verified the IEE certificate and confirmed that the project is in compliance with National Policy on Environmental and Social Sustainability of the Hydropower Sector in Lao PDR/13/.

Overall, based on the information provided, the project appears to be well-prepared and in line with the necessary legal and regulatory frameworks governing hydro power projects in Laos.

Participation under other GHG programs:

The project activity is neither registered nor applied for \in other GHG or Non GHG program(s) to obtain the certified GHG emission reductions. The project owner declared that the emission reductions claimed under this project activity will not be double counted under any other GHG program. The declaration /15/ on the same has been provided and found appropriate.

The project is not registered under any other GHG programs and is solely seeking registration with the VCS program which is confirmed through the declaration /15/ provided by the PP. Additionally, the project has not been rejected by any other GHG programs which is confirmed by the validation team by checking the various registries official project websites. Regarding other forms of credit, the project does not participate in emissions trading programs or any other mechanisms involving GHG allowance trading, nor has it sought or received any other GHG-related credits, including renewable energy certificates this has been conformed through host country emission trading policy as Lao PDR is a non-annex 1 country there were no emission trading system is in practice([chrome-extension://efaidnbnmnnibpcajpcglclefindmkaj/https://unfccc.int/sites/default/files/resource/Laos%20PDR%20final.pdf](https://efaidnbnmnnibpcajpcglclefindmkaj/https://unfccc.int/sites/default/files/resource/Laos%20PDR%20final.pdf)).For supply chain (Scope 3) emissions, there have been no public statements from the owner(s) or retailer(s) of the impacted goods or services regarding the issuance of VCUs. Similarly, the project proponent has not issued any public statements about VCUs related to the greenhouse gas emission reductions associated with the project. According to the Corporate Value Chain (Scope 3) Accounting and Reporting Standard, the project activity does not involve any direct impact emissions associated with goods or services and is therefore not related to Scope 3 emissions, rendering is inapplicable to the project.

Sustainable development contributions

The project contributes to the following Sustainable Development benefits:

SDG 7-Affordable and clean energy

The project activity contributes to generating electricity from renewable sources, that helps in country energy security, reduces the GHG emissions and encourages clean, renewable and efficient technologies.

SDG 8- Decent work & economic growth

The project is a clean technology investment in the region, which would not have taken place in the absence of the VCS benefits. In addition, the project activity will generate permanent and temporary employment opportunity within the vicinity of the project. This directly and indirectly positively effects the economy of nearby populace.

SDG 13- Climate Action

Project activity generates renewable energy-based electricity and mitigates the CO₂ emissions which would have been generated from the grid-connected power plants. Project has already commissioned and started reducing the emissions. Hence, complied to the SDG 13

The same was found reasonable based on the interviews and supporting documents submitted during the site visit.

Additional information relevant to the project:

No commercially sensitive information has been excluded from the public version of the project description as verified during the interviews.

Validation team confirms that the description in the project description is accurate, complete, and provides an understanding of the nature of the project.

3.2 Safeguards

3.2.1 No Net Harm

Being a renewable project activity, there is no negative impacts from the project activity. Hydro energy systems do not produce air pollution or greenhouse gases. In general, using hydro energy has a positive, indirect effect on the environment when hydro energy replaces or reduces the use of other energy sources that have larger effects on the environment, i.e., there is not net harm.

The project is a hydro project, there is no considerable change in the water storage area and hence no negative impact due to the project. The same is confirmed from site visit and interview with the project proponent. The same is reconfirmed from the satellite maps of before and after the project implementation. PP has clarified that there is no potential environment or socio-economic impacts is expected from the project as explained under section 2.1 of the VCS PD /02/ which is found OK. The Project activity meets all local laws and regulation of Laos PDR.

Further socio-economic impacts are rather indirect but positively related to the project activity. There are no negative socio-economic impacts from the project which is also discussed during the stakeholder interview.

The validation team confirmed that the project has no negative impact on respect for human rights and equity. The project owner adheres rigorously to non-discriminatory practices in hiring, ensuring inclusivity across race, gender, ethnicity, religion, and disability status. Furthermore, the project upholds fair labor standards, providing safe working conditions, adequate training, and protective gear, in full compliance with international human rights frameworks which is confirmed through the interview and verifying the HR policy (available at onsite) document during the site visit. Regarding property rights, the project owner possesses construction contract documents/14.12/, maintaining strict boundaries to prevent encroachment on private, stakeholder, or government property. As for ecosystem health, the project site consists mainly of secondary and disturbed forestland, with no presence of protected areas or habitats for rare or endangered species. Thus, the hydro power project demonstrates commendable efforts in safeguarding human rights, promoting equity, respecting property rights, and mitigating impacts on ecosystem health.

Validation team has interviewed the local stakeholder and confirmed that project has positive impact on the environment and proper compensation was given to people.

3.2.2 Local Stakeholder Consultation

During the onsite visit the validation team verified the local stakeholder consultation report/14.10/ which attested with the LSC photographs and attendance sheet and confirmed that the Public consultation was carried out at different levels including the Government Authorities from National, Provincial, District, Village authorities and institutions, as well as international organizations. At village level the consultation focused on the proposed affected villages which are located in upstream, downstream and project construction sites in order to provide basic information about the project. The content of public consultation includes provision of basic information about the project, its objectives and the expected potential project impacts and mitigation measures. Houaykeng village is predominantly inhabited by the Hmong-Mien ethnic group, constituting over 65% of its population, while Pakhoung's residents are mostly Lao Loum or Lao ethnic groups, comprising 63% of the village population. These communities have been resettled in the past decade, moving from approximately 3 km upstream of the original Houaykeng village under government policies aimed at relocating isolated villages closer to public services. Relocation incentives such as land user rights have facilitated this move, with some residents opting to relocate voluntarily to access government services or following new road alignments where land is available. Despite these efforts, both villages remain underdeveloped with predominantly poor economic conditions. According to the district socio-economic report of 2007-2008, Xayaboury district comprises 83 villages with 12,082 households and a total population of 67,220, including 33,919 females. The annual population growth is estimated at 2.9%. The survey of two villages indicates 167 households and 181 families, totaling 1,228 people, of whom 472 are female. The representative of each village group attended the meeting was asked to introduce the project to the other members of each village. The public consultation at village level was conducted

informally during the time of social economic survey in each impact village. The participants were from the different groups as in following:

- Representative of government official of the concerned districts;
- Representative of Lao women's union, at village level, Lao National Front and the Lao Youth organization at village level;
- Head of village and head of household;
- The socio-economic survey teams.

Summary of the consultation meeting held on 21/08/2022

Nam Houg - 1 Hydro Power project received support from all the villages around the project area. The main positive impacts of the project are as follows:

- Opportunity for employment by the Project during the construction and the operation phases;
- Opportunity to sell small goods and/or provide food for the workers at the camp;
- Benefits from an improved road-increased and all-year access;
- Availability of electricity and the opportunity to expand into small home-run businesses as well as improving everyday life; and
- Access to training and technical programs delivered by District Government (Improved access to agricultural extension services).

Post the stakeholder meeting, all the concerns raised during the meeting was properly addressed along with mitigation measures to minimise any impacts

The questions raised by the villagers are summarised in the table below:

Stakeholder Comment	Explanation (Why? How?)	Remarks
Is the air in local context polluted?	Project Participant's representative confirmed that there would be no negative permanent effect to locals during the project construction and operation phase, instead there might be only minor temporary impact due to dust emission while the mitigation measure would be adopted.	The following mitigation measures followed during the construction to avoid dust. Localized sprinkling of water to suppress the dust and suspended particles.
How is the air in local context before and during the construction?	The sources of air emission and fugitive dust will likely be from the drilling, crushing, and hauling associated with the construction activities. However, the dam construction is not directly linked to industrial processes,	The following mitigation measures followed during the

	and thus the impact of emission from specific contaminants, organic chemical compounds and particulate matters will be minimized except where the work involves welding and aerosol spraying. Furthermore, the representative mentioned that a series of mitigation measure would be prepared for the local people to improve their local air.	construction to avoid dust: Localized sprinkling of water to suppress the dust and suspended particles
Does this project affect the protection of biodiversity?	The project area is of predominantly secondary and disturbed forestland and patches of remnant woodland. There is no presence of primary forest nor are there any NPAs, PPAs or DPAs located within or nearby the Project area. These village conservation forests will, among other uses, provide habitats for wildlife and healthy forest ecosystem services. It is important for the project to make sure that threats to species will not be resulted from the presence of the construction workforce. At the same time, presence of the workforce should not result in increased pressures on the local harvest and use of NTFPs. For this reason, direct project impact on sensitive forest areas and resources will be minimal during the construction phase and negligible during operation.	Not Applicable
Will the implementation of the project help improve the working environment and quality of the workers?	The working environment of the workers will strictly implement as following measures: Promotion of preventive measures, e.g., use of condoms. Regular site clean-ups, draining water bodies that can harbour disease vector (mosquito). Conduct hygiene and sanitation awareness programmed. Support local initiatives on disease prevention campaigns. Siting fuel depot away from the living buildings and residential areas.	Not Applicable
Will the project provide jobs and generate incomes?	Project Participant's representative mentioned that all the construction works would be open for local construction company, and would request the company to recruit locally.	Not Applicable

The validation team also verified the minutes of meeting to note that no negative comments were received and the same was cross checked with the information obtained during follow up interviews with the stakeholders. Moreover, assessment team during the validation site visit also noted that a grievance register is also put onsite for the stakeholder to comment on any grievances during the operation lifetime of the project activity, the grievances from the stakeholder if found suitable will be addressed immediately by the top management and thus the approach is found appropriate for the project activity. Thus, ongoing communication of stakeholders is followed through grievance mechanism. The VVB confirms that the local stakeholder consultations have been done in accordance with VCS Standard, v4.5 requirements.

3.2.3 Environmental Impact

The project activity is expected to have positive impacts and no significant adverse environmental impacts are foreseen. Since, the project activity is an electricity generation from renewable source (i.e. Hydro energy) therefore no negative impact is envisaged. During the construction phase, there were some unavoidable minor environmental and social impacts. However, the project owner implemented effective mitigation measures to address these issues. This was confirmed during a site visit, where interviews with stakeholders and villagers showed that they were completely satisfied with the measures taken by the project proponent. All the mitigation measure taken by the project proponent were verified by the validation team during the onsite visit and found to be acceptable.

Project Phase	Environmental and Social Impacts		Mitigation Measures
	Positive	Negative	
1. Planning		Land acquisition	All acquired land was rightly compensated according to the Laws and approval by District Authorities.
2. Construction	Creation Employment for local community including services		Training and preparing skill and unskilled labor list.
		Soil Erosion from earth, rock excavation for weir, coffer and road construction	All surplus earth and stone were reused for protection construction, new road and road renovations. Planting grass and shrub in the weak soil area was done.
		Noise Impact from rock explosion and excavation	To concentrate the explosion times, explosion was avoided at noon/nighttime, using short delay blasting technology, personal protection equipment.

Project Phase	Environmental and Social Impacts		Mitigation Measures
	Positive	Negative	
		Dust pollution	Water was spraying throughout the construction activity area to keep the dust low.
		Traffic interruption	Traffic safety management, traffic warning signs.
		Impact on tree adjacent to power transmission line	Avoided tall trees during the laying of transmission line. If unavoidable, the top of trees was trimmed regularly. Workers were trained in correct techniques of tree trimming without damage to trunk or roots.
		Water pollution from weir construction	A temporary sedimentation tank with a capacity of not less than 30 m ³ was built to avoid soil discharge into the river. The existing and construction new water supply system was improved for the downstream villages.
3.Operation	Creating employment for local people, including services and tourism activities, increasing local electricity supply, decreasing fire wood		Training and preparing skill and unskilled labour list. Prepare tourism management promotion plan.
	Opportunity of water use for agriculture and aquaculture and tourism		Xayabouly District is to be promoted.
		Water flow changes under the weir, which will impact on	Downstream of the power house will not be impacted by the Project

Project Phase	Environmental and Social Impacts		Mitigation Measures
	Positive	Negative	
		aquatic ecologically downstream	
		Noise produced by machine. Domestic waste and waste water	Install soundproof doors or windows. Waste water will be treated by a small biological treatment planted underground. Some excrement will be using as organic fertilizer. Waste will collect in designated areas with anti-leakage measures (cement paved areas, awning) and then will be transferred to the garbage disposal area of the Xayabouly town that has more hygienic and safety measures.
		Waste machine oil produced by maintenance of machine and equipment.	Collect waste oil and transfer them to Xayabouly District solid waste place
		Sediment	A simple wash out facility into the structures. Sedimentation will occur at the very low load

3.2.4 Public Comments

As per section 3.18.9 of the VCS Standard, Version 4.5, all projects are subject to a 30- day public comment period. The date on which the project is listed on the project pipeline marks the beginning of the project's 30-day public comment period. The project was open for public comments on the verra website from 28/09/2023 to 28/10/2023. There are no public comments received during the GSC period.

3.2.5 AFOLU-Specific Safeguards

The project is a non-AFOLU project. Hence, this section is not required.

3.3 Application of Methodology

3.3.1 Title and Reference

Type I, Renewable Energy Projects:

"Small-scale methodology for grid-connected renewable electricity generation" – AMS I.D", Version 18.0/5/

The methodologies also refer to the latest approved version of the following tools:

- Tool 07- Tool to calculate the emission factor for an electricity system, Version 07.0/8/
- Demonstration of additionality of small-scale project activities – Version 13.1/7/
- Investment analysis, version 12.0/6/

ACM0002: Grid-connected electricity generation from renewable sources, version 21.0/5/(referred for calculation of project emissions and project density calculation as per AMS -I.D)The methodology is valid at the time of validation of the project activity have been checked and confirmed by the assessment team.

3.3.2 Applicability

The project activity applies methodology; AMS I.D. version 18.0 The applicability criteria for the methodology are assessed by the audit team by means of document review and interviews. The audit team confirms that the project activity meets the criteria of the applied methodology.

“Small-scale methodology for grid-connected renewable electricity generation” – AMS I.D”, Version 18.0

AMS I.D. version 18.0 applicability conditions	Project activity applicability	Validation Remarks
This methodology is applicable to project activities that: Install a Greenfield plant; Involve a capacity addition in (an) existing plant(s); Involve a retrofit of (an) existing operating plant(s)/unit(s); Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity is a greenfield hydro power plant substituting electricity produced on the grid. Hence, this is applicable.	The project activity consists in the installation of a new grid connected renewable power plants at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield power plant). Hence point (a) is applicable. Commissioning certificate /17/, Monthly electricity invoices /14.10/, Signed PPA /14.7/ etc. have been checked to confirm the same. Project is supplying electricity to Electricite Du Laos. PPA /14.7/ has already signed between PP and Electricite Du Laos which has been verified by the validation team. Hence the project is in compliance with latest version of the methodology AMS I.D Version 18.0.
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: a) The project activity is implemented in an existing reservoir, with no change in the volume of reservoir; b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²; c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m².	The project activity is construction of a new hydro power plant (i.e., Greenfield project) which results in a new reservoir and the power density as per definitions given in the project emissions section, is greater than 4 W/m². which is calculated as per the methodological formula of ACM0002/5/, the same is included in the section 3.3.6 of the report.	The project activity is the 15MW hydro power plant. (i.e., Greenfield project) Therefore, this criterion is applicable. Hence the project is in compliance with latest version of the methodology AMS I.D. Version 18.0.

If the new unit has both renewable and non- renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The rated capacity of the project activity is 15 MW with no provision of Co-firing fossil fuel. Hence, meeting with this criterion.	It is a renewable energy project with no provision for co-firing fossil-fuel. The project activity is the installation of a new hydro power plant. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology AMS I.D. Version 18.0.
Combined heat and power (co-generation) systems are not eligible under this category.	The project activity does not involve with co-generation. Hence it is not applicable.	The project activity is the installation of a new hydro power plant and doesn't involve any co-generation. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology AMS I.D. Version 18.0.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct¹ from the existing units.	The project activity is a new installation, it does not involve any capacity addition measures nor any replacement and hence is not applicable for the project activity.	The project activity is the installation of a new hydro power plant and not involving any capacity addition. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology AMS I.D. Version 18.0.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	The project activity does not involve retrofit, rehabilitation or replacement. Hence, this is not applicable.	The project activity is the installation of a new hydro power plant and doesn't involve any retrofit, rehabilitation measures or any replacement. Hence the project is in compliance with latest version of the methodology AMS I.D. Version 18.0.

In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS- I.C.: Thermal energy production with or without electricity” shall be explored.	This is not a landfill gas, waste gas, wastewater treatment and agro- industries projects. Hence, this criterion is not applicable.	The project activity consists of installation of a new grid connected renewable power plant at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield power plant). Hence the project is in compliance with latest version of the methodology AMS I.D. Version 18.0.
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	This criterion is not applicable as this is not a biomass-based project activity.	The project activity consists of installation of a new grid connected hydro power plant. Hence this para is not applicable. Hence the project is in compliance with latest version of the methodology AMS I.D. Version 18.0.

TOOL 07: “Tool to calculate the emission factor for an electricity system”, version 07.0

Applicability condition	Applicability to this project activity	Validation Remarks
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project activity is the installation of greenfield hydro power generation plant. “Operating margin” (OM), “build margin” (BM) and “combined margin” (CM) need to be estimated to calculate baseline emissions of the project activity that substitutes electricity in the Laos grid. Hence the tool is applicable.	This project involves electricity generation from the hydro power plant where generated electricity is exported to a grid. Thus, the applicability criterion is met. In the absence of the project activity, the equivalent amount of power would have been drawn from the Laos grid which is dominated by fossil fuel power plants. The baseline emissions are calculated from electricity supplied to the grid by the project activity multiplies with emission factor of the National grid. The emission factor calculated using OM, BM and CM using this tool. Thus, the applicability criterion is met. Justification provided by the PP is found OK. Project is supplying electricity to Laos grid. PPA /14.7/ has already signed, which has been verified by the validation team. Hence the project is in compliance with this tool.

Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off - grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 2: Procedures related to off- grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off- grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The emission factor for the project electricity system is calculated for grid power plants. Hence, this tool is applicable.	This project involves electricity generation from the hydro power plant where generated electricity is exported to a grid. Therefore, the emission factor is calculated for grid power plants. Hence the project is in compliance with this tool.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project electricity system is located totally in non- Annex I country. It is located in Lao People's Democratic Republic. Hence, the tool is applicable.	The project is located on the host country Lao PDR, which is not Annex I country, hence the criterion is applicable. Justification provided by the PP is found OK. Hence the project is compliance with this tool.
Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	There are no biofuels used in the project activity. Hence this criterion is not applicable.	This project involves electricity generation from the hydro power plant where generated electricity is exported to a

		grid. Hence the condition is not applicable.
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Tool 21: Demonstration of additionality of a small-scale project activity – version 13.1

Applicability condition	Applicability to this project activity	Validation Remarks
The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for Project participants and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	The applied methodology is not a new methodology. Project proponent has applied this tool for the demonstration of additionality in compliance with the tool. Refer to section 3.5 of PD for the detailed applicability of this tool and additionality assessment. Hence the tool is applicable.	The tool is included by an approved methodology AMS I.D version 18.0 which is the applied methodology. Thus, the application of this tool was found to be acceptable, and the applicability criterion is met. Hence the project is in compliance with this tool.

Tool 27: Investment Analysis – version 12

Applicability condition	Applicability to this project activity	Validation Remarks
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This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	The additionality of the project activity is demonstrated using “Demonstration of additionality of small-scale project activities”. Hence the tool is applicable.	The demonstration of additionality for the project activity is done using “Demonstration of additionality of small-scale project activities”. Hence, TOOL 27 is used for conducting investment analysis. Hence the project is in compliance with this tool.
In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied methodology does not contain requirements that are different from those described in this methodological tool for the investment analysis. Hence this criterion is not applicable.	The additionality for the project activity is demonstrated using “Demonstration of additionality of small-scale project activities”. The applied methodology contains requirements that are not described in this methodological tool for the investment analysis of the project activity. Hence the project is not in compliance with this tool. it is not applicable.

3.3.3 Project Boundary

As per applicable methodology AMS-I.D. Version 18, “The spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to.”

Thus, the project boundary includes the Hydro Turbine Generators and the Lao grid system.

The project does not involve any other emissions sources not foreseen by the methodologies. The greenhouse gases and emission sources included in or excluded from the project boundary are shown in table below.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

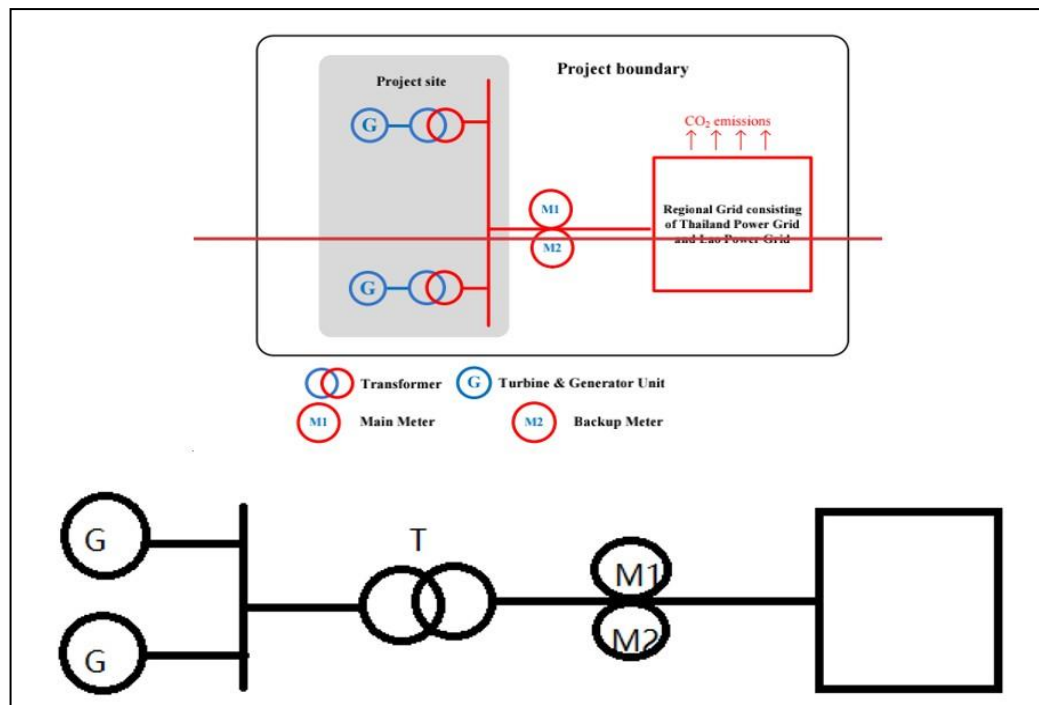


Fig. 2 Project Boundary

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Yes	Main Emission Source
	CH ₄	No	Minor Emission Source
	N ₂ O	No	Minor Emission Source

Source	Gas	Included?	Justification/Explanation
Project	CO ₂	No	Project activity does not emit CO ₂
	CH ₄	No	Project activity does not emit CH ₄
	N ₂ O	No	Project activity does not emit N ₂ O

The validation team concludes that the project boundary is correctly identified in accordance with the methodology.

3.3.4 Baseline Scenario

According to para 24 of version 18.0 of AMS I.D /06/, for the project activity including the installation of a Greenfield grid-connected renewable power plant, the baseline scenario is the following:

“If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system.

The project aims to generate electricity by using hydroelectric power to supply the increasing national electricity demand in a cleaner and sustainable manner. It reduces the air pollution caused by the grid connected power plants which are mostly fossil fuel fired. The project is a greenfield hydropower plant installation in Xayabouly province, Lao PDR. Total installed capacity of the project is 15MW. The same has been verified from FSR /14.3/, PPA /14.7/, Equipment purchase agreement/contract /14.14/, Commissioning certificates /17/ and Photos etc. In the absence of the project activity, the equivalent amount of power would have been drawn from the Lao Power Grid. Hence, the baseline for the project activity is the equivalent amount of power from the Lao power grid.

The baseline scenario selected is in compliance with all applicable legal and regulatory requirements as the implementation of project activity is a voluntary initiative and is not mandatory or a legal requirement. Based on the sectoral expertise of the validation team, the selection of baseline scenario by the project participant is more appropriate and acceptable.

The emission factor of the grid used for the project activity is as follows:

Parameter	Value	Nomenclature
EF _{grid,CM,y}	0.4504 tCO ₂ /MWh	Combined margin CO ₂ emission factor for the project electricity system in year y
EF _{grid,OM,y}	0.4504 tCO ₂ /MWh	Operating margin CO ₂ emission factor for the project electricity system in year y

EF _{grid,BM,y}	0 tCO ₂ /MWh	Build margin CO ₂ emission factor for the project electricity system in year y
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Validation team confirm that project boundary and selected sources, sinks and reservoirs are justified for the project.

3.3.5 Additionality

As per the Law on Electricity (amended version) No. 03/NA, dated 20/12/2011/22/ implementation of the renewable energy project is not mandated and are entirely a voluntary action. The project activity does not fulfil the criteria of positive list as provided in CDM Tool 32: “Methodological Tool – Positive List of Technologies” and hence additionality of the project activity is demonstrated through a project specific additionality test.

As indicated in PD, project start date and investment decision date (i.e. date of investment decision, as in the sense of CDM methodology) is identified as 24/09/2016, i.e. the signing of EPC contract/14.8/. Validation team examined thoroughly the provided additionality assessment within the VCS PD and the supplementary “Equity IRR.” workbook.

Project developer had demonstrated that the financial returns of the VCS project activity would be insufficient to justify the required capital investment as per CDM Validation and Verification Standard for project activities In the final VCS PD version 03/1/. PP has adopted a conservative approach to identify the benchmark for the project activity. The project is earning revenue from the installation of the project activity. Thus, simple cost analysis (Option I) is not appropriate. Also, in the absence of the project activity grid electricity would have been the obvious choice for the Project which requires no investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity. Therefore, benchmark analysis (Option III) is used for the project activity as per project type and decision-making context. Therefore, the Expected return on equity is considered appropriate benchmark. Accordingly, the post-tax Equity IRR has been considered as the relevant financial indicator for the project activity which is acceptable to the assessment team. Moreover, the financial indicator selected by the PP is correct based on the fact that tool do not restrict the PP to either use project IRR or Equity IRR. This is under the prerogative of the PP to select appropriate indicator based on his preferences to know the IRR based on his equity investment or debt investment. The same is thus acceptable to the assessment team. Assessment team however checked the Equity IRR calculation and found that input assumptions used for the calculation of Equity IRR are applicable at the time of investment decision of the project and thus is in accordance with the relevant guideline of the tool.

As per Methodological tool “Investment Analysis” Version 12, “In situations where an investment analysis is carried out in nominal terms and the available IRR benchmarks are in real terms, project participants shall convert the real term values of benchmarks to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information is also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five years after the start of the project activity shall be used”.

The investment analysis has been carried out in Nominal terms. Accordingly, default value has been adjusted by adding 5 year forecasted inflation rate published by IMF. This is found to be appropriate as the central bank of Thailand does not provide long term inflation forecast or inflation target.

Appendix A in Methodological tool “Investment Analysis” Version 12 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Laos = 20.68%

The Required return on equity (benchmark) was computed in the following manner: Nominal Benchmark = $\{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$

Where:

Default value for Real Benchmark = 20.68% (as per Appendix of Methodological tool “Investment Analysis” Version 12)

Inflation Rate: 2.9 % as per the 5 year forecast provided in IMF’s World Economic Outlook database: April 2016 for Lao. The source (<https://www.imf.org/en/Publications/WEO/weo-database/2016/April/weo-report?c=544,&s=PCPI,PCPIPCH,&sy=2017&ey=2021&ssm=0&scsm=1&scs=0&ssd=1&ssc=0&sic=0&sort=country&ds=.&br=1>) has been verified and found to be available at the time of investment decision and correct.

Nominal Benchmark estimated = $(1 + 20.68\%) * (1 + 2.90\%) - 1 = 24.18\%$

The period considered for Post Tax Equity IRR calculations is 30 years, which corresponds to the operational lifetime of the project activity.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Appropriateness of the input parameters:

The input parameters in the financial analysis have been taken as per the values and assumptions applicable and available at the time of decision to invest in the project activity in line with Paragraph 10, investment analysis tool version 12.0.

The post-tax equity IRR for the project activity at the time of investment decision comes out to 9.92%. Verification team assessment of all the input parameters is as follows:

Particulars	Value	Unit	Source/Remarks
Capacity of the project	15	MW	Verified against FSR/14.3/ which was available at the time of investment decision and cross verified against PPA/14.7/ and commissioning certificate /17/of the project. Further, the same has been confirmed during onsite visit.

Annual Net generation	87.280	GWh	Verified against FSR/14.3/ which was available at the time of investment decision. As the FSR is prepared by third party engineering company (The Exploration and design institute under sinohydro bureau 14 co.,Ltd ,on Jan 2016), it is found to be in line with EB 48, Annex 11. The actual generation achieved in the last one year is 87.280 GWh/14.2/ which equal to the value considered in the FSR. Hence, validation team confirms that the PLF considered for the project activity is conservative; hence acceptable.
Project cost	45186.20	k USD	Verified against FSR/14.3/ which was available at the time of investment decision. The cost is reasonable with the market rate during investment decision time which varies from 4300 kUSD/MW to 4600 kUSD/MW. The project cost considered in the IRR calculation is found to be appropriate. The verification team also crosschecked the project with the project cost considered in other projects submitted under various carbon mechanisms which are commissioned: From the above cost, it is found that the project cost considered for the project activity is on par with the project cost of other similar projects. Also the PP's project completion report/14.11/ is verified and found that the property & plant cost of the project is 45186.20 k USD which is same as the project considered for the FSR. However, even with the actual project cost, the IRR works out to be less than the benchmark.
Debt	80%	%	The debt equity ratio is based on the FSR/14.3/ which was available at the time of investment decision. Normally, the infrastructure projects are given loan between 60 to 80% of the project cost. Hence the debt equity ratio considered is acceptable
Equity	20%	%	

Interest rate	10.00%	%	The interest rate is based FSR/14.3/ which was available at the time of investment decision. As verified from PDDs of other registered projects, the interest rate considered in this project is found that this is also in line with the interest rate considered by other hydro projects in Laos.
Debt Repayment tenure	12	years	Loan Tenure is based on the FSR/14.3/ which was available at the time of investment decision. As verified from PDDs of other registered projects, the loan tenure considered in this project is found that this is also in line with the interest rate considered by other hydro projects in Laos.
Moratorium	0	year	
Repairs (% of annual income)	5%	%	The O&M cost and its escalation is based on the FSR/14.3/ which was available at the time of investment decision. The total O&M cost is works out to be 10.5 k USD. As verified from the company financials, the actual O&M cost for the year 2021 is 283 k USD which is much higher than the O&M cost considered in the IRR calculation sheet. Hence, the O&M cost considered by PP is found to be conservative & appropriate.
Insurance (% of fixed asset)	75.31033	k USD	
Transmission line maintenance	6.632	K USD	
O&M Labour	10.5	k USD	
Tariff	0.065	USD/kWh	The tariff base rate is based on the FSR/14.3/ which was available at the time of investment decision. This is also crosschecked through the PPA/14.7/ signed for the project activity with grid authority. Hence, the tariff considered in the IRR calculation is found to be appropriate. As verified from the tariff amendment the actual tariff is flat 0.065 USD/kWh for entire years. With this actual tariff, the IRR further goes down. Hence, the tariff considered for the investment decision is more conservative.
Depreciation Rate (Book)	4.00%	%	The book depreciation is based on the FSR/14.3/ available at the time of investment decision. This is found to be appropriate considering the life time of 30 years.
Salvage Value	10%	%	The salvage value of is based on the assumption calculation Also, this is found to be standard in the market. Hence, the salvage value considered is found to be appropriate

Income tax rate for year 1 to year 10	10	%	The salvage value of is based on the assumption calculation as per IRR/3/. This is also found to be actual income tax rate applicable at the time of investment decision. Hence, the income tax rate value considered for IRR calculation is found to be appropriate is conformed through the taxation certificate/14.15/.
Income tax rate for year 11 to year 30	24		

Financial calculation and conclusion

The equity IRR calculations were provided in a spreadsheet. The calculation was verified and found to be correct by project validation team; as well as the assumptions used in the calculation were deemed to be correct. The equity IRR without carbon credit revenues is 9.92% which confirms that the proposed project activity in absence of the carbon credit benefits and compared to the benchmark return on equity 24.18% is not financially attractive.

Sensitivity Analysis:

A sensitivity analysis has been carried out for parameters contributing more than 20% revenues and costs, to demonstrate the robustness of the financial analysis. The parameters for which sensitivity analysis done are annual power generation, change in tariff, project costs, operational and maintenance cost, interest rate and debt equity ratio. Sensitivity analysis was conducted for $\pm 10\%$ variation. Reasonable variations for these parameters were checked by calculating the variation necessary to reach the benchmark and then discussing the likelihood for that to happen.

Variation %	-10%	Normal	10%	Variation required to reach benchmark	Value to breach benchmark
Tariff	7.83%	9.92%	12.09%	59.50%	0.104 USD/kWh
Generation	7.83%	9.92%	12.09%	59.50%	139.212 GWh
Project Cost	12.30%	9.92%	8.05%	-37.70%	28151.00 k USD
O&M Cost	10.07%	9.92%	9.77%	-852.30%	-2829.42 k UDS

Conclusion

The results of sensitivity analysis show that even with a variation of $\pm 10\%$ in tariff, PLF, project cost, and O&M cost, equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favorable conditions. Hence, the financial barrier is found to be valid. Hence, the proposed project shall be considered as additional.

3.3.6 Quantification of GHG Emission Reductions and Removals

Assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline Emissions:

As per para 22 of the applied methodology AMS I.D. Ver 18.0, baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where:

BE_y	Baseline emissions in year y (tCO ₂ /yr).
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{grid,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”.

As per paragraph 23 of the applied methodology, the grid emission factor is calculated in a transparent and conservative manner as follows

- A combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the “Tool to calculate the emission factor for an electricity system”; or
- The weighted average emissions (in t CO₂/MWh) of the current generation mix. The data of the year in which project generation occurs must be used.

The Project Participant has selected option a for calculation of emission factor for the project activity which is appropriate as per methodological requirement.

Calculation of EG_{PJ}

Because the project activity is the installation of a new grid-connected renewable power plant/unit at a site where no renewable power plant was operated prior to the implementation of the project activity, then:

$$EG_{PJ,y} = EG_{facility,y} \text{ (as per para 25 of the applied methodology)}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Therefore, the baseline emissions are calculated as follows:

$$BE_y = EG_{facility,y} \times EF_{grid,CM,y}$$

Calculation of $EF_{grid,CM,y}$

Full calculation process of the grid emission factor in line with the “Tool to calculate the emission factor for an electricity system”, version 07.0 has been presented in an ER spread sheet.

4KES assessment is given below:

- Simplified CM is applied since the project activity is located in a LDC country. Under the simplified CM, the operating margin emission factor ($EF_{grid,OM,y}$) must be calculated using the average OM (Option (d) in Step 3 of the tool).
- The average OM emission factor is calculated as the average emission rate of all power plants serving the grid, using the methodological guidance as described under Step 4 (section 6.4.1) of the tool for the simple OM, but also including the low-cost/must-run power plants in all equations.
- “Ex ante option” is chosen for the project. The emission factor is calculated using a 3-year (2020-2022) generation-weighted average, based on the most recent data available at the time of submission of the PD to the DOE for validation.

For simple OM method, Option B *‘based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system’* is selected in the PD and is justified as data for single power station and power unit are not publicly available in Lao PDR. 4KES confirms that the calculation of emissions from total fuel consumption in ER spreadsheet is correct.

Following the simplified CM approach applied to the proposed project, $W_{BM}=0$ and $W_{OM}=1$ are correctly applied in the renewed PD. Thus, $EF_{grid,CM,y} = EF_{grid,OM,y}$.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$$

Where:

$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (tCO₂/MWh)

$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (tCO₂/MWh)

WOM = Weighting of operating margin emissions factor (%) = 100% (for simplified CM)

WBM = Weighting of build margin emissions factor (%) = 0% (for simplified CM)

Therefore,

$$EF_{grid,CM,y} = 0.4504 + 0 = 0.4504 \text{ tCO}_2/\text{MWh}.$$

Grid emission factor value to calculate the emission reductions of the hydro power plant project is 0.4504 tCO₂/MWh.

Hence the baseline emission factor is calculated as below:

$$BE_y = 87,280 \times 0.4504 = 39,309 \text{ tCO}_2$$

Further Tool to calculate the emission factor for an electricity system (version 07.0) is used for this emission factor calculation which is in compliance with the applied methodology i.e. AMS I.D Version 18.0 /05/. Hence the selected emission factor for the project activity is accepted by the validation team.

Project Emissions: -

According to para 39 of AMS I.D. Version 18.0, the project emissions for hydro power plants to be considered following the procedure described in “ACM0002: Grid-connected electricity generation from renewable sources”.

Therefore, as per para 31 of ACM0002 Ver 21.0, project emissions are calculated using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

PE_y	Project emissions in year y (tCO ₂ e/yr)
$PE_{FF,y}$	Project emissions from fossil fuel consumption in year y (tCO ₂ /yr)
$PE_{GP,y}$	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO ₂ e/yr)
$PE_{HP,y}$	Project emissions from water reservoirs of hydro power plants in year y (tCO ₂ e/yr)

As per para 37 of the methodology ACM0002 Ver 21.0, the power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

PD = Power density of the project activity (W/m^2)

Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W) (ie, 15,000,000 W)

Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero

A_{PJ} = Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m^2) (ie, as per the satellite map it is 159,835 m^2)

A_{BL} = Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m^2). For new reservoirs, this value is zero.

Hence, $PD = (15,000,000 - 0) / (159,835 - 0) = 9.87 W/m^2$

As per ACM0002, version 21.0 methodology, If the power density of the project activity is greater than 4 W/m^2 and less than or equal to 10 W/m^2 , $PE_{HP,y}$ is calculated

$$PE_{HP,y} = \frac{EF_{Res} \times TEG_y}{1000}$$

Where:

$PE_{HP,y}$ = Project emissions from water reservoirs (t CO₂e/yr)

EF_{Res} = Default emission factor for emissions from reservoirs of hydro power plants (kg CO₂e/MWh) -As per ACM0002, EF_{Res} table in page number 13, it is 90 kgCO₂e/MWh

TEG_y = Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y (MWh) – as per FSR, it is 87,280 MWh per year

Hence, $PE_y = (90 \times 87,280) / 1000 = 7,856 t CO_2e/yr$

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Leakage Emissions: -

According to para 42 of AMS I.D. Version 18.0, no leakage emissions are considered.

Hence, $LE_y = 0$.

Emission reductions (ER_y)

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e)

BE_y = Baseline emissions in year y (t CO₂e)

PE_y = Project emissions in year y (t CO₂e)

LE_y = Leakage emissions in year y (t CO₂)

Validation team confirm that the methodological choices/approaches and found OK. Validation team has verified the ex-ante ER calculation /02/ and found OK. The ex-ante Emission reduction is summarized below.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
01-September-2023 to 31-December-2023	13,139	2,626	0	10,513
01-January-2024 to 31-December-2024	39,309	7,855	0	31,453
01-January-2025 to 31-December-2025	39,309	7,855	0	31,453
01-January-2026 to 31-December-2026	39,309	7,855	0	31,453
01-January-2027 to 31-December-2027	39,309	7,855	0	31,453
01-January-2028 to 31-December-2028	39,309	7,855	0	31,453
01-January-2029 to 31-December-2029	39,309	7,855	0	31,453

01-January-2030 to 31-August-2030	26,170	5,230	0	20,940
Total	275,161	54,986	0	220,174
Total number of crediting years	7			
Annual average over the crediting period	39,309	7,855	0	31,453

The emission reduction is 220,174 tCO₂e is a emission reduction of first crediting period (of the renewable type) is confirmed by the validation team.

Through the validation process, 4KES has confirmed that:

- All the assumptions and data used by the project participants are listed in the PD including their references and sources;
- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the PD;
- All values used in the PD are considered reasonable in the context of the VCS project activity;
- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, leakage and emission reductions;
- All estimates of the baseline emissions can be replicated using the data and parameter values provided in the PD.

3.3.7 Methodology Deviations

The project activity has used the AMS.I. D, ver.18.0/5/ the PP has completely followed the methodology for the development of the VCS PD. The validation team has confirmed that the PP does not make any deviations from the applied methodology.

3.3.8 Monitoring Plan

The GHG validator has assessed the monitoring plan, which outlines a comprehensive approach to data collection, monitoring, and record maintenance in line with AMS I.D Version 18.0/5/. The plan addresses all critical aspects, including project management, emergency preparedness, personnel training, and QA/QC procedures. It specifies that data will be collected and recorded by a dedicated team, with technical oversight provided by a technical manager. Monitoring will involve continuous tracking of electricity input and output through standardized meters, with proper calibration/14.4/ and verification against invoices/14.10/. The plan includes protocols for handling non-conformities, ensuring data integrity through regular equipment calibration, and maintaining thorough records. Emergency procedures are in place to handle unforeseen events, and data will be archived for two years post-crediting period. Overall, the monitoring plan demonstrates a robust framework for ensuring reliable data collection and accurate GHG emission reduction calculations.

Data and parameters available at validation

Following parameters are fixed for the entire crediting period and are mentioned in section 5.1 of the VCS PD.

Data / Parameter	EF _{grid,0M,y}	VVB assessment
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Data unit	tCO ₂ /MWh	The ex-ante parameter is as per the applied methodology AMS I.D. (Version 18.0). The applicable grid emission factor value to calculate the emission reductions of the hydro power plant project is 0.4504 tCO₂/MWh. Validation team has checked the standardized baseline and found applicable for the project activity. Further Tool to calculate the emission factor for an electricity system (version 07.0)/8/ is used for this emission factor calculation which is in compliance with the applied methodology i.e., AMS I. D. Hence the selected emission factor for the project activity is accepted to the Validation team.
Description	Operating Margin CO ₂ emission factor in year y	
Source of data	Calculated	
Value applied:	0.4504	
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07” as 3-year generation weighted average using data for the years 2018, 2019 & 2020.	
Purpose of Data	Baseline Emission calculation	
Comments	This parameter is fixed ex-ante for the entire crediting period	

Data / Parameter	EF _{grid,BM,y}	VVB assessment
Data unit	tCO ₂ /MWh	The ex-ante parameter is as per the applied methodology AMS I.D. (Version 18.0). The applicable grid emission factor value to calculate the emission reductions of the hydro power plant project is 0 tCO₂/MWh. Validation team has checked the standardized baseline and found applicable for the
Description	Build Margin CO ₂ emission factor in year y	
Source of data	Calculated	
Value applied:	0	

Justification of choice of data or description of measurement methods and procedure applied	Calculated as per “Tool to calculate the emission factor for an electricity system, version 07”	project activity. Further Tool to calculate the emission factor for an electricity system (version 07.0) is used for this emission factor.
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Data / Parameter	EF _{grid,CM,y}	VVB assessment
Data unit	tCO ₂ /MWh	The ex-ante parameter is as per the applied methodology AMS I.D. (Version 18.0). The applicable grid emission factor value to calculate the emission reductions of the hydro power plant project is 0.4504 tCO ₂ /MWh.
Description	Combined Margin CO ₂ emission factor in year y	
Source of data	Calculated	
Value applied:	0.4504	

Justification of choice of data or description of measurement methods and procedures applied	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y} = EF_{grid,OM,y} * WOM + EF_{grid,BM,y} * WBM$ Where: $EF_{grid,BM,y}$ = Build margin CO2 emission factor in year y (tCO2/MWh) $EF_{grid,OM,y}$ = Operating margin CO2 emission factor in year y (tCO2/MWh) WOM = Weighting of operating margin emissions factor (%) = 100% (for simplified CM) WBM = Weighting of build margin emissions factor (%) = 0% (for simplified CM)	Validation team has checked the standardized baseline and found applicable for the project activity. Further Tool to calculate the emission factor for an electricity system (version 07.0) is used for this emission factor calculation which is in compliance with the applied methodology i.e., AMS I. D. Hence the selected emission factor for the project activity is accepted to the Validation team.
Purpose of Data	Calculation of baseline emissions	
Comments	This parameter is fixed ex-ante for the entire crediting period	

Data / Parameter	EF _{Res}	VVB assessment
Data unit	kgCO ₂ e/MWh	The emission factor (EF Res) of 90 kgCO₂e/MWh for emissions factor from reservoirs, as specified in ACM0002: Grid-connected electricity generation from renewable sources V.21.0/5/ was verified and confirmed by
Description	Default emission factor for emissions from reservoirs	
Source of data	ACM0002: Grid-connected electricity generation from renewable sources V.21.0	

Value applied	90 kgCO ₂ e/MWh	the validation team as a default value established by the standard for calculating emissions associated with reservoirs in hydroelectric power projects. This fixed value was used for power density calculations throughout the entire crediting period and did not require specific measurement or justification beyond the default setting. It was essential to ensure that this default emission factor aligned with the most recent guidelines and research to maintain accuracy in reporting and compliance.
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable	
Purpose of Data	Power density calculation	
Comments	This parameter is fixed ex-ante for the entire crediting period	

Data / Parameter	Cap_{BL}	VVB assessment
Data unit	W	The parameter Cap_{BL} , representing the installed capacity of the hydro power plant before the project activity, was verified and confirmed by the validation team as correctly applied at 0 W for new hydro power plants, in accordance with ACM0002: Grid-connected electricity generation from renewable sources V.21.0./5/ This value was appropriate as it accurately reflects there were no projects present prior to this project. Since this parameter is fixed ex-ante for the entire crediting period and used for power density calculations, no additional measurement or justification beyond the default setting was necessary. This practice was confirmed to be in line with standard procedures, ensuring
Description	Installed capacity of the hydro power plant before the implementation of the project activity. For new hydro power plants, this value is zero	
Source of data	ACM0002: Grid-connected electricity generation from renewable sources V.21.0	
Value applied	0	
Justification of choice of data or description of measurement methods and procedures applied	Not Applicable	
Purpose of Data	Power density calculation	
Comments	This parameter is fixed ex-ante for the entire crediting period	

		consistency in reporting and validation.
Data / Parameter	ABL	VVB assessment
Data unit	m ²	The parameter ABL, which represents the area of the reservoirs measured in square meters when full prior to the project activity, was verified and confirmed by the validation team as correctly set to 0 m ² for new reservoir constructions. This value was consistent with the methodology, which dictates a zero value for new reservoirs to indicate no pre-existing reservoirs. As this parameter is fixed ex-ante for the entire crediting period and used in project emission calculations, its application was in accordance with the methodology/5/, ensuring precise reporting.
Description	Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m2). For new reservoirs, this value is zero	
Source of data	Project site	
Value applied	0	
Justification of choice of data or description of measurement methods and procedures applied	Since this is project involves construction of new reservoirs, it is considered as zero as per methodology.	
Purpose of Data	Project Emission calculation	
Comments	This parameter is fixed ex-ante for the entire crediting period	

Data and parameters monitored

Following parameters are monitored during the crediting period and are mentioned in section 5.2 of the VCS PD:

Data / Parameter	EG PJ,y	VVB assessment
Data unit	MWh/y	The validation team has validated and confirmed that the data parameter EG PJ,y, which represents the quantity of net electricity generated by the project plant and supplied to the grid, is robustly
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh	
Source of data	Electricity meter readings	

Description of measurement methods and procedures to be applied	The difference of final value of export and import is used for monthly values of net electricity supplied to the grid by the project activity and same value will be considered for ER calculations.	monitored and measured. This data, recorded in MWh/year, is obtained from electricity meter readings using a methodology where net electricity supplied is calculated as the difference between export and import values, with continuous monthly recording. The monitoring equipment, including energy meters with an accuracy class of 1.0 for kWh and 2.0 for kvarh, adheres to a five-year calibration schedule /14.4/and is managed by EDL, ensuring stringent QA/QC procedures. The Energy meter make – DTSD341
Frequency of monitoring/recording	Continuous and monthly measurement	
Value applied	87,280 MWh	
Monitoring equipment	The electricity exported / supplied by the plant to pooling substation and further to substation. This meter also measures electricity imported by the plant from the grid. Energy meter with an accuracy: class 1.0 for kWh, class 2.0 for kvarh Energy meter make – DTSD341 Serial Number: 1. Main – 250318494, Backup – 250318495 2. Main – 250318493, Backup – 219014741	Serial Number: Main – 250318494, Backup – 250318495 Main – 250318493, Backup – 219014741 the details are confirmed the calibration certificate/14.4/ and the same is reflected during the site visit also. Monthly data is cross-verified with sale invoices/14.10/, and conservative guidelines are followed in cases of delayed meter calibration/14.4/. The data is used for baseline emission calculations and will be archived for two years after the crediting period or the last issuance of VERs.
QA/QC procedures to be applied	The frequency of calibration of energy meters is once in 5 years. The monthly electricity supplied/exported by the project activity in the JMR report is cross checked with the monthly invoices of sale. In the absence or delay in the meter calibration appropriate Guidelines will be applied to confirm the conservativeness of metering. The metering arrangement, accuracy class of meters, calibration frequency is under control of EDL and PP does not have any control on it.	
Purpose of data	Calculation of baseline emissions	
Calculation method	Net electricity supplied to the grid by the project plant in a given month = Export(kWh)– Import (kWh)	

Comments	Data will be archived in paper & electronic form for two years after the end of crediting period or of the last issuance of VERs for this project activity, whichever occurs later.	
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Data / Parameter	TEGy	VVB assessment
Data unit	MWh/year	The validation team has reviewed the data parameter TEGy, which represents the total electricity produced by the project activity, encompassing both grid-supplied and internally used electricity. The data, measured in MWh/year, is obtained from continuous electricity meter readings/14.1/ at the project activity site, with monthly recording(JMR)/14.2/. The applied value is 87,280 MWh, the continuous monitoring ensures accurate tracking of the total electricity produced.
Description	Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads, in year y	
Source of data	Project Activity Site	
Description of measurement methods and procedures to be applied	Electricity meters	
Frequency of monitoring/recording	Continuous measurement and at least monthly recording	
Value applied	87,280	
Monitoring equipment	Electricity meters	
QA/QC procedures to be applied	NA	
Purpose of data	Calculation of project emissions	
Calculation method	NA	
Comments	NA	

Data / Parameter	Cap _{PJ}	VVB assessment
Data unit	W	The validation team has reviewed the data parameter Cap _{PJ} , which denotes the installed capacity of the hydro power plant implementation.
Description	Installed capacity of the hydro power plant after the implementation of the project activity	

Source of data	Project Site	The data, recorded in watts (W), is sourced from the project site and is determined based on manufacturer's specifications/18/, commissioning data/17/.The value applied is 15,000,000 W, with measurement performed once at the beginning of each crediting period.
Description of measurement methods and procedures to be applied	Determine the installed capacity based on manufacturer's specifications or commissioning data or recognized standards	
Frequency of monitoring/recording	Once at the beginning of each crediting period	
Value applied	15000000	
Monitoring equipment	NA	
QA/QC procedures to be applied	NA	
Purpose of data	Calculation of project emissions	
Calculation method	NA	
Comments	NA	

Data / Parameter	A _{PJ}	VVB assessment
Data unit	m ²	The validation team has reviewed the data parameter A _{PJ} , during the onsite visit and confirmed the same with cross verifying the FSR/14.3/ and EPC contract/14.8/ which represents the area of the reservoir(s) measured in square meters (m ²) when full, following the implementation of the project activity. This data is obtained from the project site and determined through topographical surveys, maps, or satellite images. The applied value is 1,520,000 m ² , with measurement conducted once at the beginning of each crediting period.
Description	Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full	
Source of data	Project Site	
Description of measurement methods and procedures to be applied	Measured from topographical surveys, maps, satellite pictures, etc	
Frequency of monitoring/recording	Once at the beginning of each crediting period	
Value applied	1520000	
Monitoring equipment	-	

QA/QC procedures to be applied	-	
Purpose of data	Calculation of project emissions	
Calculation method	-	
Comments	-	

3.4 Non-Permanence Risk Analysis

Non-permanence risk analysis is not applicable as the Project is not an AFOLU (Agriculture, Forestry and Other Land Use) project.

4 VALIDATION OPINION

Kosher Climate India Private Limited has commissioned '4K Earth Science Private Limited' to do VCS validation of the project activity: "Nam Houng - 1 Hydro Power Project" having VCS ID 4588, with regard to the relevant requirements of VCS Standard version 4.5 /4/; to attain real, measurable, additional and permanent emission reductions.

A risk-based approach has been followed to perform this validation. In the course of the validation, 02 Corrective Action Request (CAR) and 03 Clarification Request (CL) and 00 Forward Action Request (FAR) were raised and successfully closed.

The validation is based on the VCS PD, additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and supporting documents made available to the validation team by project proponent.

As a result of the validation, the validation team confirms that:

- The project fulfils criteria of VCS Standard Version 4.5.
- The project is in line with all relevant VCS requirements.
- The project baseline is sufficiently justified in the PD.
- The calculation of the emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of is 220,174

tCO₂e /2/ is most likely to be achieved within the 07 years of renewable crediting period.

- No restrictions or uncertainties were identified related to the validation.

Validated GHG emission reductions and removals in the period:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
01-September-2023 to 31-December-2023	10,513
01-January-2024 to 31-December-2024	31,453
01-January-2025 to 31-December-2025	31,453
01-January-2026 to 31-December-2026	31,453
01-January-2027 to 31-December-2027	31,453
01-January-2028 to 31-December-2028	31,453
01-January-2029 to 31-December-2029	31,453
01-January-2030 to 31-August-2030	20,940
Total estimated ERs	220,174
Total number of crediting years	07
Average annual ERs	31,453

APPENDIX 1: DOCUMENT REFERENCES

The following table lists the documentation that were reviewed during this validation of the project activity:

1.	VCS PD for “Nam Houng - 1 Hydro Power Project version 1.0 dated 11/09/2023 VCS PD for “Nam Houng - 1 Hydro Power Project version 2.0 dated 23/01/2024 VCS PD for “Nam Houng - 1 Hydro Power Project version 3.0 dated 15/07/2024 VCS PD for “Nam Houng - 1 Hydro Power Project version 4.0 dated 04/09/2024
2	Emission Reduction Calculation Excel sheet (ex-ante) and version 01 dated 11/09/2023 Emission Reduction Calculation Excel sheet (ex-ante) and version 03 dated 15/07/2024
3	VCS: VCS Program Guide, version 4.4 dated 29.08.2023 VCS Program Definitions (v4.3, dated 21/12/2022) VCS Registration & Issuance Process (v4.3, dated 17/01/2023)
4	VCS: VCS Standard, version 4.5 dated 04.10.2023 VCS Validation and Verification Manual version 3.2 dated 19/10/2016
5	“Small-scale methodology for grid-connected renewable electricity generation” – AMS I.D Version 18.0 ACM0002: Grid-connected electricity generation from renewable sources V.21.0
6	Investment analysis, version 12.0
7	Demonstration of additionality of small-scale project activities – Version 13.1
8	Tool 07- Tool to calculate the emission factor for an electricity system, Version 07.0
9	VERRA: Project search: https://registry.verra.org/app/search/VCS/All%20Projects CDM project standard for project activities, version 03.0 CDM validation and verification standard, version 03.0
10	https://registry.verra.org/app/projectDetail/VCS/4588

11	VERRA Rules and Requirements: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#rules-and-requirements
12	VCS: VCS Project Description template, VCS Version 4.2 of 21/12/2022
13	National Policy on Environmental and Social Sustainability of the Hydropower Sector in Lao PDR https://policy.asiapacificenergy.org/sites/default/files/National%20Policy%20on%20the%20Environmental%20and%20Social%20Sustainability%20of%20the%20Hydropower%20Sector.pdf
14	/14.1/ENERGY METRE SPECIFICATION /14.2/JOINT METRE READING /14.3/FEASBLITY STUDY REPORT(FSR) PREPARED BY ((THE EXPLORATION AND DESIGN INSTITUTE UNDER SINOHYDRO BUREAU 14 CO.,LTD ,ON JAN 2016),) /14.4/METRE CALIBRATION REPORT /14.5/IEE REPORT AND CERTIFICATE DATED 16/02/2023 /14.6/ELECTRICITY LICENSE BY SEPRATE ENERGY AND MINERAL RESOURCE DEPARTMENT ON 30/12/2016 /14.7/POWER PURCHACING AGREEMENT(PPA) BETWEEN STATE ELECTRICITY COMPANY OF LAOS AND NAM HUNG 1 HAI COMPANY DATED 3RD AUGUST 2020 /14.8/EPC CONTRACT SIGNED BETWEEN NAM HOUNG-1 HPP CO LTD AND ZHEEJIANG JINLUN E&M CO LTD DATED 24/09/2016 /14.9/ THE LOCAL STAKEHOLDER CONSULTATION (INDIVIDUAL INVITATIONS, PHOTOGRAPHS, ATTENDANCE SHEETS ETC.) /14.10/SALES INVOICES /14.11/ PROJECT COMPLETION REPORT date FEB 2023 /14.12/ CONSTRUCTION CONTRACT BETWEEN PHON TAN VILLAGE XAYSETHA DISTRICT, VIENTAINE,LAO PDR AND XENOHYDRO COMPANY NO.14 IN YUNNAN PROVINCE,CHINA DATED 05/12/2013 /14.13/ PROJECT ALLOTMENT LICENSE DATED 11/05/2015 /14.14/THE EQUIPMENT PURCHASE CONTRACT BETWEEN 1 MAY GROUP COMPANY LIMITED AND NAM HOUNG-1 HPP CO LTD /14.15/ TAXATION CERTIFICATE

15	Declaration to comply with various requirements under VERRA Declaration form on avoidance of double counting of the project activity
16	Emission Factor for LAO PDR National Grid
17	Project Commissioning Certificate
18	Technical details of metre
19	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
20	2006 IPCC Guidelines for National Greenhouse Gas Inventories; 5th Assessment Report
21	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 4 Energy, Chapter 10 Introduction
22	Law on Electricity (amended version) No. 03/NA, dated 20/12/2011; https://policy.asiapacificenergy.org/sites/default/files/Electricity%20Law.pdf
23	Environmental Protection Law (2013 Ed.) (Law No. 29/NA of 2012) https://policy.asiapacificenergy.org/sites/default/files/Environmental%20Protection%20Law%202013%20Revised%20Ed%20%28Lao%29.pdf
24	Decree on Environmental Impact Assessment dated January 31, 2019 https://data.laos.opendevlopmentmekong.net/en/laws_record/decreed-on-environmental-impact-assessment

APPENDIX 2: ABBREVIATIONS

CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
EB	Executive Board
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse Gases
IPCC	Intergovernmental Panel for Climate Change
ISO	International Organization for Standardization
LE	Local Expert

MP	Monitoring Period
MR	Monitoring Report
MW	Mega Watt
MWh	MegaWatt hour
PD	Project Description
PP	Project proponent
QA/QC	Quality Assurance/Quality Control
tCO ₂	Tonnes of Carbon Dioxide
TE	Technical Expert
TL	Team Leader
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit

APPENDIX 3: VALIDATION FINDINGS

TABLE 01 CLs from this Project Validation

CL ID	01	Section no.	DOCUMENTS	Date: 12/01/2024
Description of CL				
Project Participant's (PP) is requested to submit the following documents / supporting's: 1. Proof for Start date of project. 2. Calibration certificates for the energy meters 3. The project ownership proof 4. The equipment purchasing contract. 5. Proof for claimed SDG. 6. LOA 7. PPA				
Project Participant's response				Date: 19/01/2024

1. COD certificate has been provided as a proof for start date of project. 2. The project is commissioning on 01st September 2023. Calibration of the energy meters will be done once in 5 years and the certificate will be provided during verification. 3. Power Purchase Agreement and EPC are submitted as proof for ownership for the project. 4. The equipment purchasing contract is provided. 5. Proof for claimed SDGs will be provided during verification. 6. LOA is not applicable as this is a VCS project. VCS communication agreement is shared for your reference 7. PPA is provided as supporting document.	
Documentation provided by Project Participant's	
1. COD certificate 2. PPA 3. EPC 4. VCS communication agreement	
DOE assessment	Date: 03/02/2024
The given documents are found to be acceptable, Hence CL01 is closed.	

CL ID	02	Section no.	3	Date:12/01/2024
Description of CL				
1. In section 3.4 The given link is not accessible to find the inflation rate kindly revise and give the exact link. 2. The reference number for footnote link 8 is nowhere in the page 23 kindly clarify.				
Project Participant's response				Date: 22/01/2024
1. The reference number is revised and provided to find the inflation rate under section 3.4 of the PDD 2. The reference number for footnote link 8 is provided under "Default value for Real Benchmark = 20.68%" in the page 23 of the revised PD				
Documentation provided by Project Participant's				
1. Updated PD				
DOE assessment				Date: 03/02/2024
2.Kindly include the updated version foot link of tool 27,the link which you provide does not have the default value of benchmark for Lao. 3.In many place of the PD it is mentioned as 'expected to be commissioned' kindly revise. Hence CL02 is open				
Project Participant's response				Date: 05/02/2024

The footlink is included and the required revision is done.	
Documentation provided by Project Participant's	
DOE assessment	Date:05/02/2024
The given source and explanation are found to be ok hence CL03 is closed	

CL ID	03	Section no.	4	Date: 12/01/2024
Description of CL				
1. Kindly provide the source link of operating margin and build margin. 2. Kindly give the source link of EF_{Res} for reference.				
Project Participant's response				Date: 24/01/2024
1. The emission factor is calculated using the data from EGAT annual report and EDL Electricity statistic report. The source link is provided in the ER sheet 2. The EF_{Res} is taken from the default value provided in the methodology ACM002, version 21.0. The same is provided in section 4.2 of the PD				
Documentation provided by Project Participant's				
1. Updated PD				
DOE assessment				Date: 03/02/2024
The given source and explanation are found to be ok hence CL03 is closed.				

TABLE 02 CARs from this Project Validation

CAR	01	Section no.	4	Date:12/01/2024
Description of CAR				
1. Kindly include the ACM0002 methodology in the applicability condition and give justification. Kindly review the emission reduction calculation as w_{OM} = Weighting of operating margin emission factor (%) and w_{BM} = Weighting of build margin emission factor (%) were seems to be incorrect. 2. Kindly submit the proof document or link for Default emission factor for emissions from reservoirs of hydro power plants, as the value has no source its directly taken into account of ER project emission calculation.				
Project Participant's response				Date: 25/01/2024

1. The project does not use ACM002. Hence the applicability condition is not provided. Since we are using simplified combined margin, the WOM and WBM are considered as per para 90 of Tool 07 v7.0 2. The default emission factor is taken from the data / parameter table 6, page no. 24 provided in the methodology ACM002, version 21.0. The same is provided in section 4.2 of the PD	
Documentation provided by Project Participant's	
1. Updated PD	
DOE assessment	Date: 03/02/2024
The explanation given by PP is acceptable, Hence CAR01 is closed.	

CAR	02	Section no.		Date:12/01/2024
Description of CAR				
1. Kindly provide the source and evidence for $EG_{PJ,y}$ Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh.				
Project Participant's response				Date: 25/01/2024
$EG_{PJ,y}$ is based on the electricity generation estimation provided in the FSR and the same is included in the PD				
Documentation provided by Project Participant's				
1. Updated PD				
DOE assessment				Date:
The explanation and revision done in PD are found to acceptable, Hence CAR02 is closed.				

Table 3. FAR from this validation

FAR ID	01	Section no.		Date:
Description of FAR				
Project participant response				Date: DD/MM/YYYY
Documentation provided by project participant				
DOE assessment				Date: DD/MM/YYYY

APPENDIX 4: TEAM COMPETENCE

<u>Certificate of Competence</u>						
Name	☒ Mr. ☐ Ms.	Ma Paa Puratchikkanaal				
Qualification Procedure	Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GHG Projects.					
Appointed to work as:						
	CDM Validator/Verifier	Team Leader	Team Member	Technical Expert	Technical Reviewer	Financial Expert
<i>Appointed</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Appointed Date</i>	15-07-2023					
Authorized to work as Technical Expert for:						
<i>Authorized Technical Area</i>	Sectoral Scope		TA Code	Technical Area within the scope		
	Energy industries (renewable - / non-renewable sources)		1.1	Thermal energy generation		
	Energy industries (renewable - / non-renewable sources)		1.2	Renewables		
	Energy demand		3.1	Energy demand		
	Construction		6.1	Construction		
	Waste handling and disposal		13.1	Solid waste and wastewater		
	Waste handling and disposal		13.2	Manure		
	Agriculture		15.1	Agriculture		
Authorized to work as Local Expert for:						
<i>Country/Countries</i>	India and Sri Lanka					
<u>Compliance check by:</u> Anand S. R.						

<u>Certificate of Competence</u>		
Name	☒ Mr. ☐ Ms.	Gurubalaji K
Qualification Procedure	Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GHG Projects.	

Appointed to work as:						
	CDM Validator/Verifier	Team Leader	Team Member	Technical Expert	Technical Reviewer	Financial Expert
<i>Appointed</i>	Yes	No	Yes	Yes	No	No
<i>Appointed Date</i>	15-07-2023					
Authorized to work as Technical Expert for:						
<i>Authorized Technical Area</i>	Sectoral Scope	TA Code		Technical Area within the scope		
	Energy industries (renewable - / non-renewable sources)	1.2		Renewables		
Authorized to work as Local Expert for:						
<i>Country/Countries</i>	India					
Compliance check by: Anand S. R.						

<u>Certificate of Competence</u>						
Name	☒ Mr. ☐ Ms.	Chetan Swaroop Sharma				
Qualification Procedure	Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GHG Projects.					
Appointed to work as:						
	CDM Validator/Verifier	Team Leader	Team Member	Technical Expert	Technical Reviewer	Financial Expert
<i>Appointed</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Appointed Date</i>	15-07-2023					
Authorized to work as Technical Expert for:						

<i>Authorized Technical Area</i>	Sectoral Scope	TA Code	Technical Area within the scope
	Energy industries (renewable - / non-renewable sources)	1.1	Thermal energy generation
	Energy industries (renewable - / non-renewable sources)	1.2	Renewables
	Energy distribution	2.1	Energy distribution
	Energy demand	3.1	Energy demand
	Waste handling and disposal	13.1	Solid waste and wastewater
	Waste handling and disposal	13.2	Manure
Authorized to work as Local Expert for:			
<i>Country/Countries</i>	India		
Compliance check by: Anand S. R.			

<u>Certificate of Competence</u>											
Name	☒ Mr. ☐ Ms.	Mr. Phoutkham Singphchon									
Qualification Procedure	Fulfil the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GHG Projects.										
Appointed to work as:											
	CDM Validator/Verifier	Team Leader	Team Member	Technical Expert	Technical Reviewer	Financial Expert					
<i>Appointed</i>	No	No	No	No	No	No					
<i>Appointed Date</i>	15-07-2023										

Authorized to work as Technical Expert for:			
<i>Authorized Technical Area</i>	Sectoral Scope	TA Code	Technical Area within the scope
Authorized to work as Local Expert for:			
<i>Country/Countries</i>	Lao People's Democratic Republic		
<u>Compliance check by:</u> Anand S. R.			